

MCS 481 Project One : planar convex hulls and maps
due Wednesday 23 September 2026 at noon

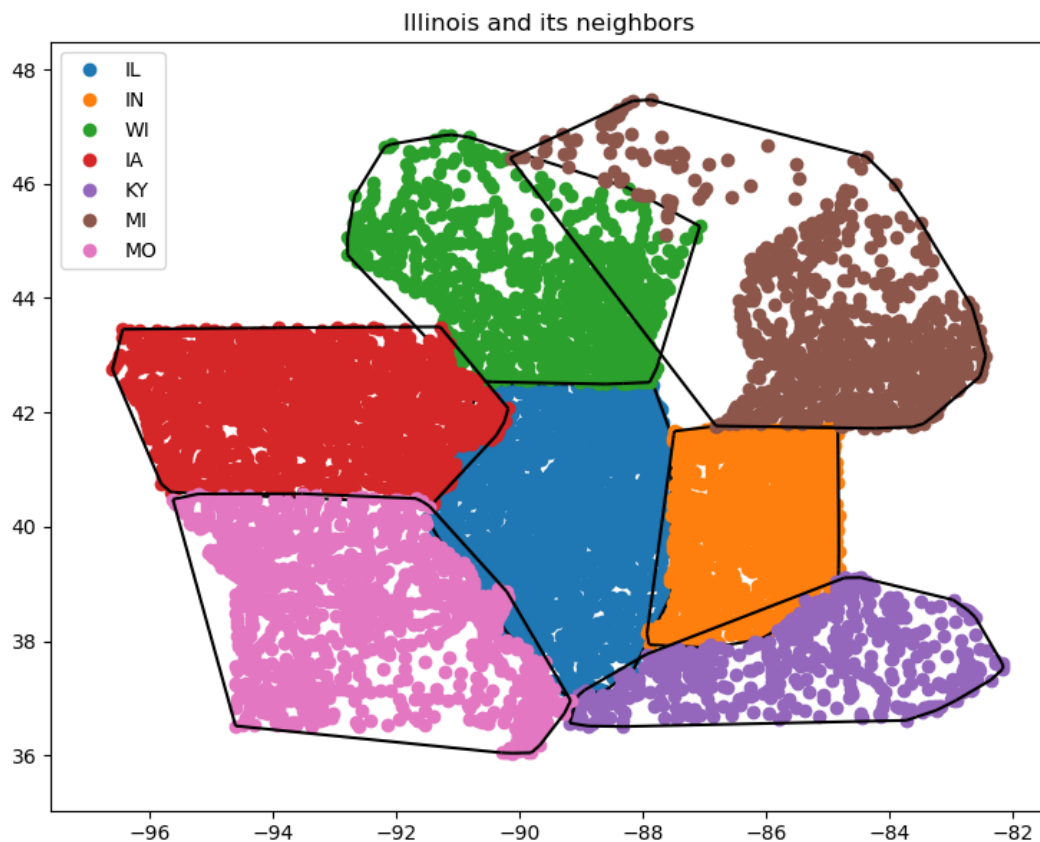
The goal of this first project is to apply the planar convex hull algorithms to draw maps of places in Illinois and its neighboring states.

1. Convex Hulls of Places in the Midwest

All places within the United States can be downloaded from

[https://data.census.gov/table/GEOINFO2025.GEOINFO?g=010XX00US\\$1600000_040XX00US17](https://data.census.gov/table/GEOINFO2025.GEOINFO?g=010XX00US$1600000_040XX00US17)

in the file `GEOINFO2025.GEOINFO-Data.csv` which contains of each place the name, state, latitude and longitude. Using longitude and latitude as the (x, y) -coordinates of the places for the state of Illinois, and its six neighboring states, the convex hulls are computed and plotted as shown below.



The `ConvexHull` and `convex_hull_plot_2d` of `scipy.spatial` were used to make the above plot.

Assignment One. Download the coordinates of all places of Illinois and its neighboring states to make the above plot.

For every state report the number of vertices of the convex hulls.

2. Fixing State Borders

The plot on the previous page approximates the borders of the states but obviously, the convex hulls do not accurately represent the state borders.

Assignment Two. Using the convex hulls of the places in Illinois and each of its neighboring states, design an algorithm to adjust the border so that the border lines correctly separate places from each state. If a place in Indiana happens to lie at the wrong side of the edge of the convex hull of Illinois, then one way to correct the border is the replace that edge by two line segments which contain the place.

1. What is the cost of adjusting the border lines, as function of the number of places?
2. Demonstrate the running of your algorithm by drawing the map with corrected borders.

3. Doubly Connected Edge Lists

If the map of Illinois and its neighbors is stored as a doubly connected edge list, then we can walk around each of the states.

Assignment Three. Define the tables with vertex, edge and face records to store the map as a doubly connected edge list.

Illustrate the use of this data structure to walk around the borders of the state. For each place on the border visited on such walk, print the coordinates and the name of the place.

4. the Deadline is Wednesday 23 September 2026 at noon

Upload your answers to gradescope before noon on Wednesday 23 September 2026.
Your solution consists of

1. The specification of your computers and the version of the software used.
2. The code for the programs you used in the experiments.
3. Tables with properly formatting of the essential characteristics for each experiment: the dimension of input and output, and the running time. Raw data may be provided as an appendix.
4. Your interpretation of the results and conclusions.

You may work in pairs for this project.

If you work in a pair, then one member of the pair must email the name of the partner to janv@uic.edu before 5pm on Wednesday 16 September.

If you have questions or difficulties with the project, feel free to ask questions during office hour.